

Analysis of Instant Ramen Ratings, Brands, Varieties, and Packagings Worldwide

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Overview

Instant ramen is enjoyed by people all over the world, especially college students. It has endless flavor varieties for an affordable, quick, and easy to prepare foodstuffs. In fact, in the year 2000, Japan named instant ramen the country's best invention of the 20th century (ahead of the Sony Walkman.)¹ That is why understanding the data surrounding ramen is so interesting; there are multiple flavors and packagings of ramen produced by countries on almost every continent of the world. Different cultures may find the access to the variety of flavors accommodating, making it palatable for almost everybody. The globalization of instant ramen has made profound impacts on populations in times of cost budgeting distresses, making it a reliable food staple.² Unfortunately, there is not much aggregated data online that tells us the story of instant ramen. Thankfully, a review site dedicated to rating thousands of types of instant ramen made from all around the world would help assist in telling us that story.

I found the data set from Kaggle.com, an online community that publishes various datasets over a variety of categories. This report is from the "Ramen Ratings" dataset located on Kaggle.com (<https://www.kaggle.com/datasets/residentmario/ramen-ratings>) that pulls from a ramen rating website called Ramen Rater. Ramen Rater has over 2,500 instant ramen reviews — dating back to 2009 — with data that covers a wide range of various brands, flavors, packagings, star ratings, review numbers, and countries of origin.

In this report, we will answer five different questions by using data analysis. We will examine ratings by flavors, the top five brands with the most varieties, the mean, median, and mode ratings of the varieties from these top five companies, what the average ratings the top rated 100 ramens are by countries who produce 100 or more varieties, and the most common form of packaging styles of instant ramen for countries.

¹ <https://abcnews.go.com/International/story?id=819468&page=1>

² <https://www.theguardian.com/world/2023/nov/20/cost-of-living-crisis-fuels-global-appetite-for-instant-ramen>

A First Glimpse of Our Dataset

We will take a first glimpse at the dataset for us to determine a few things before we dive in:

1. We want to understand how the data is formatted and written
2. We want to understand the property values
3. We wish to learn about the ratings given for the ramen

	Review #	Brand	Variety	Style	Country	Stars	Top Ten
0	2349	Fantastic	Noodles Chicken & Corn Flavour	Cup	Australia	3	NaN
1	2308	Fantastic	Noodles Chicken Chow Mein Flavour	Cup	Australia	4	NaN
2	2301	Suimin	Noodle With Chicken Flavour	Cup	Australia	3	NaN
3	2296	Suimin	Noodles With Chicken Flavour	Cup	Australia	3	NaN
4	2285	Fantastic	Noodles Beef Flavour	Cup	Australia	3	NaN
5	2274	Suimin	Noodles With Braised Beef Flavour	Cup	Australia	4	NaN
6	2253	Singa-Me	Instant Noodles Mi Goreng Flavour	Cup	Australia	4	NaN
7	2231	Fantastic	Noodle Chicken Flavour	Cup	Australia	4	NaN
8	2219	Fantastic	Noodles Crispy Bacon Flavour	Cup	Australia	1	NaN
9	2188	Singa-Me	Instant Noodles Chicken Flavour	Cup	Australia	2.5	NaN

(Source: from author)

Above is a screen capture of the dataset of the ramen-ratings.csv file. Using `data.head(10)` we are shown the first 10 rows with seven columns: Review #, Brand, Variety, Style, Country, Stars, and Top Ten. Stars ultimately refer to the rating system (from 0 being the least to 5 being the best) for each ramen variety in its respected row. Later in this report, we will clean and shape the data to fit our analysis.

Aside from this, we also wanted to get an idea of the total size of our data set. We used `data.info()` to pull a more detailed examination of what types of data we have within our set.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2580 entries, 0 to 2579
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Review #    2580 non-null  int64
 1   Brand       2580 non-null  object
 2   Variety     2580 non-null  object
 3   Style       2578 non-null  category
 4   Country     2580 non-null  object
 5   Stars       2577 non-null  float64
 6   Top Ten     41 non-null    object
dtypes: category(1), float64(1), int64(1), object(4)
memory usage: 123.9+ KB
```

(Source: from author)

From this information, we've gathered that there are 2,580 reviews, brands, varieties, and countries. However, some data is missing from styles and stars which will account for later.

Preparing & Cleaning Our Data

In order to present an accurate analysis, our data preparation will need to clean any inconsistencies, irrelevant data, or missing values by deleting them. We will need to delete the Top Ten column as it is unnecessary for our findings. Next, we will clean the 'Stars' column by removing non-numeric characters using `data['Stars'] = data['Stars'].str.replace(r'[^d.]+', '')` and also convert to a numeric value while coercing errors to a NaN value using `data['Stars'] = pd.to_numeric(data['Stars'], errors='coerce')`

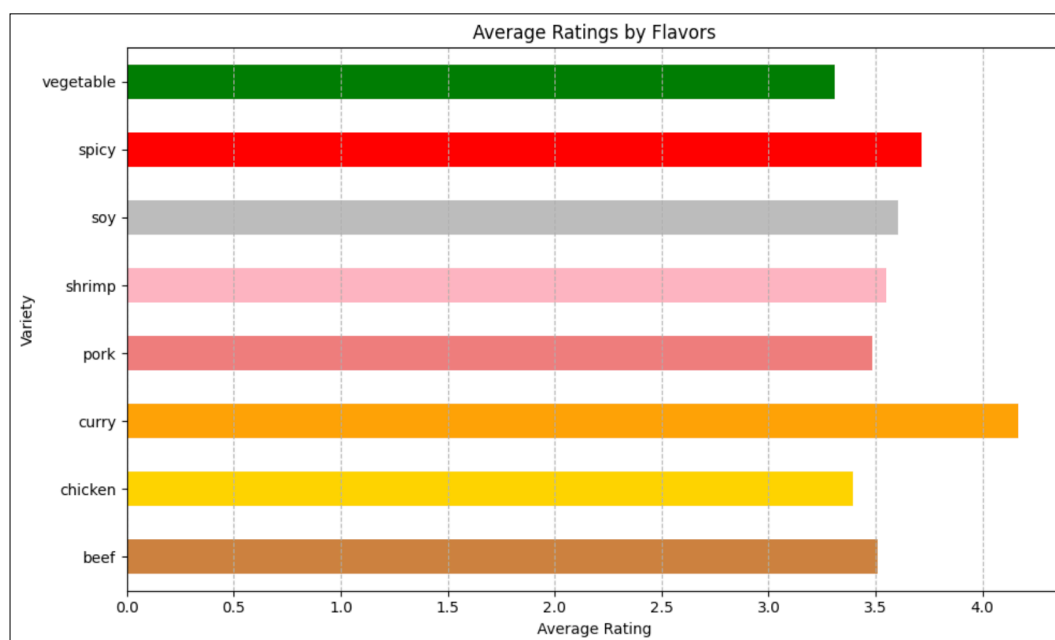
Analyzing Our Data

With our data preprocessing complete through data cleaning and transformation, we are now ready to begin our data analysis.

Our first question was this: Of the 2,580 ratings, what are the most popular flavors?

We started by creating a Python list of the most common instant ramen flavors:

```
flavors = ['beef', 'pork', 'chicken', 'shrimp', 'spicy', 'curry', 'vegetable', 'soy']
```



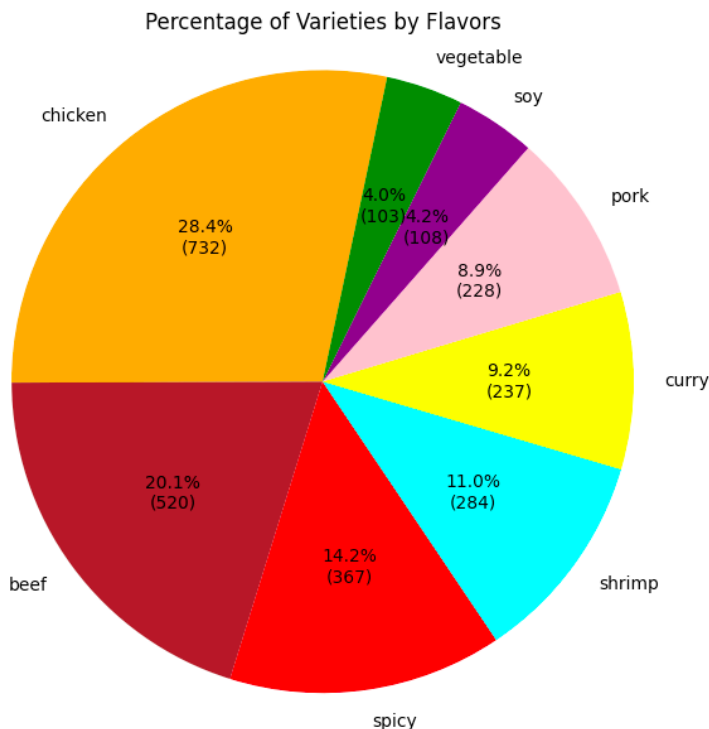
(Source: from author)

The most widely eaten meat in the world is pork, then poultry, beef, and goat/sheep.³

It is fascinating to see that curry was the most popular flavor, more so than both beef and the chicken (a meat that is generally more culturally accepted.)

³ <https://ask.usda.gov/s/article/What-is-the-most-consumed-meat-in-the-world>

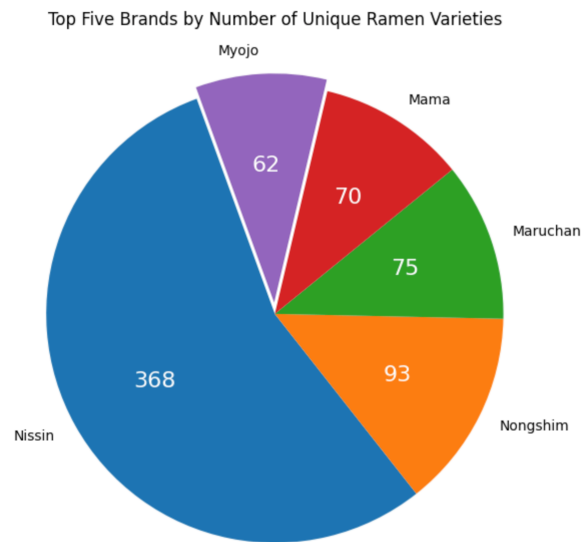
It is important to note that the ratings for flavor varieties are affected by the percentage of their total make up of 2,580 variety categories. In order to gain a better understanding of how many flavors we have, we will add up the their percentages into a pie chart.



(Source: from author)

A fascinating aspect of the Percentage of Varieties by Flavors is in the disparity between ratings and representation. Although curry scores the highest average rating of 4.4, it accounts for only 9.2% of varieties, totaling 237 flavor options. Comparatively, chicken flavor varieties make up 28.4% (732) of the total, averaging a rating of 3.4. The wide distribution suggests a diverse scale of ratings within this category. Also interesting to note is the lower rating with the vegetable flavors of ramen, scoring at a 3.25 yet making up only 4.0% (103) of the varieties.

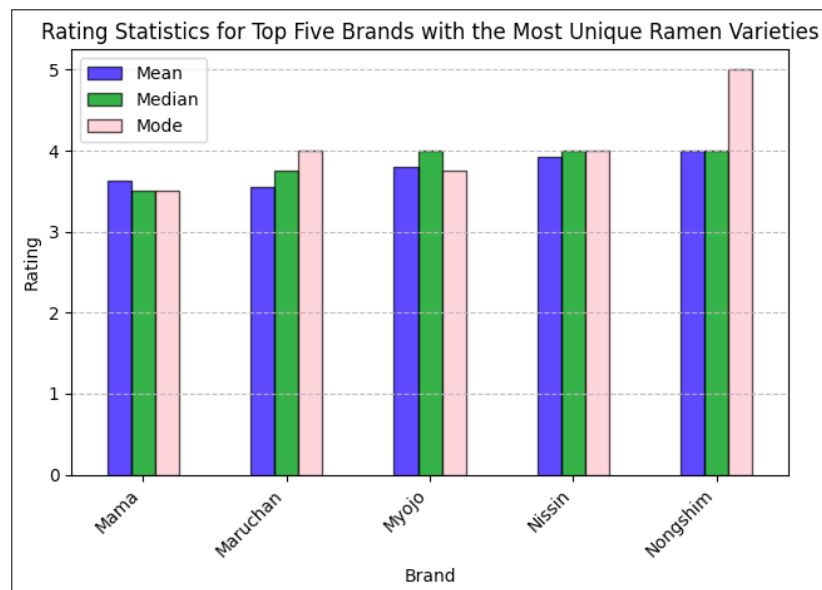
The next question we wanted to explore was what were the top five brands that had the most amount of unique ramen varieties. We designated the top five companies by the amount of ramen varieties that they have produced. We were also curious to find out if there were any sort of findings we can derive from these companies about whether or not their market share was partially the result of their flavor ratings, which we will explore later.



(Source: from author)

Nissin Foods was the founding company of Top Ramen, which first invented instant ramen noodles in 1958. Given this history, it makes sense why Nissin scored number one in the companies with the most flavor varieties. In second place is Nongshim, a South Korean food company that introduced its famous Shin Ramyun in 1986. The third company with the most varieties is Maruchan, who first produced instant ramen in 1961 and is Nissin's direct instant ramen competitor in Japan. In fourth place is Mama noodles, Thailand's flagship instant noodle company founded in 1972. Finally, Myojo, having made instant ramen products in the 1990's and providing 62 different varieties, became a wholly owned subsidiary of Nissin foods in 2006. From these findings, with the exception Nongshim, there seems to be a correlation between the years of which these companies first produced instant ramen to the amount of varieties they have.

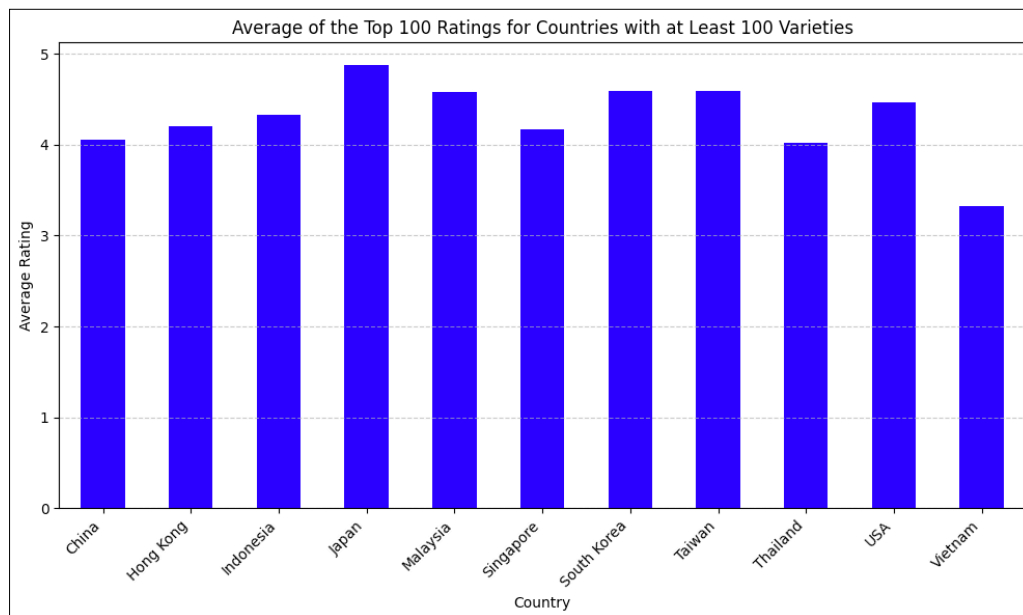
Our third question sought to discover what the average, middle, and most frequent ratings were among the instant ramen varieties made by these top five companies.



(Source: from author)

This analysis showed us the comparisons between each company in terms of their ratings' average, frequency distribution, and median value. We were able to gain deeper insights of the relative performance and popularity of these instant ramen products from each brand. Interestingly enough, Nongshim scored the highest ratings when accounting for the mode despite being the second most instant ramen variety producing company.

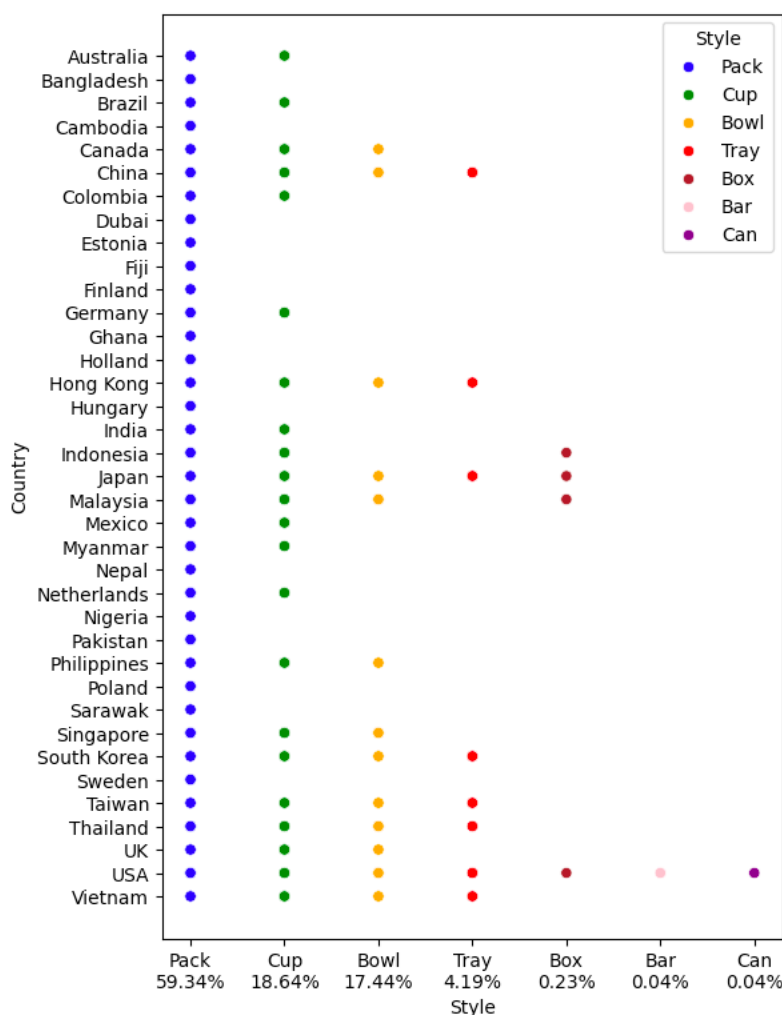
Looking at the top brands alone for their ratings wasn't enough. It was easy to get tunnel vision with just rating the most popular brands simply due to their market share of their vast instant ramen varieties. Instead, our fourth question wondered what the average rating was for countries who had at least 100 or more different varieties of ramen made. We also wanted to include their top 100 rated ramens to gain a better understanding of their best rated varieties.



(Source: from author)

Our insights provided us with surprising results: the United States of America had a competitive 4.5 rating despite not having a top five instant ramen producing company. South Korea, whose Nongshim company represented the second top instant ramen brand was slightly above with a 4.6 rating alongside Taiwan and Malaysia, who also did not have top five variety producing companies.

For our final question, we wondered how countries packaged their instant ramen. We assumed, due to cost effectiveness, that packs (plastic wrapping) was the most common. In fact, our data showed that the more complex and presumably expensive the packaging was, the less likely countries were to use it.



(Source: from author)

Our prediction was correct in determining the pack style to be the most utilized among all of the countries listed in our dataset. Interestingly enough, only the United States of America used the more complex styles of packagings like bars and cans.

Conclusion

The world of instant ramen is fascinating to explore. We can see among the 2,580 ratings of the vast varieties of ramen flavors, both curry and spicy scored the highest in taste, even surpassing beef, chicken, and pork despite cumulatively making up more than half of the measured ramen flavors. This insight could help brands in producing combinations of spicy and/or curry instant ramen flavors in order to potentially increase sales.

Predictively enough, the top five brand companies of ramen were represented by the amount of varieties that they had, along with their historical footing within the instant ramen industry.

Nissin and Nongshim, the top two instant producing companies, consistently showed high ratings for their varieties when accounting across the measures of central tendencies. For countries who made 100 or more varieties and only counting their top 100 ratings, we found it interesting how competitive varieties were made in Malaysia, Taiwan, and the USA when compared to the countries whose instant ramen companies made up the top five producing brands. Also fascinating was seeing that at the time of the data curation, there were a total of 38 different countries each producing their own instant ramen varieties. Everywhere around the world from countries like Poland to Fiji have some sort of instant ramen product. And we can clearly see that, possibly due to economic costs, packs were the most used packaging style for instant ramens across all countries.